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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, DENIM, COTTON, SHRUNK AND UNSHRUNK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, denim, cotton, shrunk and unshrunk.

1.2 Classification.

1.2.1 Types, classes and styles. The cloth shall be of the following types, classes and styles (see 6.2):

Type I - 8.0 or 8.8 ounces.
Type II - 9.0 or 9.8 ounces.
Type III - 9.9 or 11.0 ounces.
Type IV - 11.0 or 12.3 ounces.

Class 1 - Unshrunk.
Class 2 - Shrunk.

Style A - Whiteback.
Style B - Colored or tinted.

2. APPLICABLE DOCUMENTS

FSC 8305

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2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specification, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act (16 CFR 303)

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article approval. The requirement for First Article will be as specified by the procuring activity (see 6.2).

3.3 Material.

3.3.1 Yarn. The yarn shall be made of cotton which has been thoroughly cleaned, carded, drawn and evenly spun into singles or ply yarns for both the warp and filling.

3.4 Color. The color shall be as specified and shall match the standard sample when available (see 3.1, 6.2 and 6.3). Style A, whiteback cloth shall contain dyed warp yarns and white filling yarns. When Blue 3301 shade is specified (see 6.2 and 6.4), the shade shall be obtained by the use of vat or indigo dyestuffs. The filling yarns may be either blended or tinted. Style B cloth shall contain dyed warp yarns and dyed or tinted filling yarns.

3.4.1 Matching. When a standard sample is available, the dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness. Unless otherwise specified (see 6.2), the dyed cloth shall show colorfastness to light (after 40 hours exposure), laundering (after 3 cycles) and dry crocking, equal to, or better than, the standard sample when tested as specified in 4.5. When no standard sample is available, or when the standard sample is not a reference for colorfastness, the dyed cloth shall show "good" colorfastness to light (after 40 hours exposure), and "fair" fastness to laundering and dry crocking.

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3.5 Physical requirements. The finished cloth shall conform to the minimum requirements specified in table I when tested as specified in 4.5.

TABLE I - Physical requirements

Type	Class	Weight per square yard	Yarns per inch		Breaking strength	
			Warp	Filling	Warp	Filling
		Ounces			Pounds	Pounds
I	1	8.0	62	39	145	53
	2	8.8	63	42	145	58
II	1	9.0	66	43	150	65
	2	9.8	67	46	150	70
III	1	9.9	68	42	167	70
	2	11.0	69	45	167	75
IV	1	11.0	70	42	177	85
	2	12.3	71	45	177	90

3.6 Weave. Unless otherwise specified, the weave shall be a 3-harness construction, 2 up and 1 down right-hand twill (see 6.2).

3.7 Width. The width inclusive of selvages shall be as specified (see 6.2).

3.8 Nonfibrous materials. The starch and protein content including chloroform soluble and water soluble material of the finished cloth shall not exceed the percentage for the applicable type as shown when tested as specified in 4.5.

Type	Percent
I	14
II, III, and IV	13

3.9 Shrinkage or stretch. Class 2 cloth shall be preshrunk and shall not have a residual shrinkage or stretch of more than 2.0 percent in either direction of the warp or filling when tested as specified in 4.5. The preshrinking process used shall not be identified by name or trademark, either on the cloth or the ticket or package.

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3.10 pH. The pH of the cloth shall be not lower than 5.5 nor more than 8.5 when tested as specified in 4.5.

3.11 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces or on full width rolls as specified in PPP-P-1134.

3.12 Fiber identification. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act (see 2.1).

3.13 "Face" identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece or roll.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. When required, the first article of the finished cloth shall be visually inspected for shade, finish and appearance and shall be tested for the chemical and physical properties in accordance with the applicable test methods specified in 4.5.

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated.

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4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.4.

4.3.2.1 Yard-by-yard examination. Each piece or roll in the sample shall be examined on the face side only. When the total yardage in the piece or roll does not exceed 100 yards, the entire yardage in the piece or roll shall be examined. When the total yardage in the piece or roll exceeds 100 yards, only 100 yards shall be examined. Defects as defined in Section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.1.1 except that only those slubs and knots which exceed the maximum limit shown in figure 1 of Fed. Std. No. 4 shall be scored. No continuous linear yard or increment of one yard on the measuring device of the inspection machine from any one piece or roll within the sample, shall be penalized more than 4 points. The sample unit for this examination shall be one linear yard. The sample size shall be 20 pieces or rolls selected from 20 containers For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards exceeds the following:

40.0 points.

For determination of acceptability on the individual piece or roll quality basis, the lot shall be unacceptable if two or more individual pieces or rolls exceed the following:

55.0 points.

If no individual piece or roll exceeds the point level, the lot shall be accepted. If one piece or roll exceeds the point level, a second sample of 20 pieces or rolls shall be selected for individual piece or roll quality examination only. If no piece or roll in the second sample exceeds the point value, the lot shall be accepted. If one or more pieces or rolls exceeds the value, the lot shall be rejected. Point computation for the lot quality and individual piece or roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

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4.3.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

NOTE: The following defects when present shall be scored four points for each yard in which they occur:

Width of cloth less than specified.
Excessive neppiness.
Poor dye penetration, mottled, streaky or cloudy.

4.3.2.2 Examination for length.

4.3.2.2.1 Individual pieces or rolls. During the yard-by-yard examination, each piece or roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces or rolls in the sample are defective in respect to length.

4.3.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces or rolls in the sample is less than the total of the lengths marked on the tickets.

4.3.2.3 Examination for style and shade. During the yard by yard examination, each piece or roll in the sample shall be examined for style (A and B) and shade. Any piece or roll in the sample that is the incorrect style, offshade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.3.2.4 Examination for "face" identification, identification of preshrinkage process and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each piece or roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces or rolls in the sample have "face" identification missing from either or both ends, contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labelled, ticketed, or invoiced in accordance with the Textile Fiber Products Identification Act.

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4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine whether packaging, packing and marking comply with Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable and as listed in table II, shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determination made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3 1/2 continuous yards full width of the finished cloth. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II - Test methods

Characteristics	Requirement paragraph	Test method
Yarn identification	3.3.1	1200 1/
Dyestuff identification	3.4	Std Cmcl 1/
Colorfastness to:		
Light	3.4.2	5660
Laundering	3.4.2	5610 2/
Dry crocking	3.4.2	5651
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Weave	3.6	Visual 3/
Nonfibrous material	3.8	2611
Shrinkage or stretch:		
Warp	3.9	5550
Filling	3.9	5550
pH	3.10	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ Only the stain on the cotton fibers of the color transfer cloth shall be evaluated.
3/ One determination per sample unit and the results reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A anc C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-1134.

5.2 Packing. Packing shall be A, B, or C (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is primarily intended for use in the manufacture of clothing items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Type, class and style required (see 1.2.1)
(c) Whether first article is required (see 3.2)
(d) Color of cloth required (see 3.4)
(e) When colorfastness required is other than specified (see 3.4.2)
(f) When weave required is other than specified (see 3.6)
(g) Width required (see 3.7)
(h) When length is other than specified (see 3.11)
(i) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

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6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Note. A suggested but not mandatory dyestuff formulation for Blue 3301 is the use of Vat Blue 43 for the warp yarns and Denivat Everblack-W blended with natural cotton for the filling yarns.

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